

Tossaporn (Tree) Saengja

Experienced in developing advanced deep learning models
(generative AI, LLMs) and leading research over five years.

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education

Massachusetts Institute of Technology

Master of Engineering in Computer Science and Engineering 4.8/5.0 2019-2020

Bachelor of Science in Computer Science and Engineering 4.8/5.0 2015-2019

International Olympiad in Informatics 2013 (IOI), Bronze, 85th among 299 contestants (77 countries)

ACM Boston Preliminary 2016 (BOSP), 4th place among 37 teams (16 schools)

Teaching Assistant for MIT 6.824: Distributed Systems (Prof. Robert Morris, Spring 2020)

experience

SiData+, Siriraj Hospital — Data Scientist 2024-Present

- Developing a novel medical reasoning Large Language Model (LLM) with SCB 10x, focusing on improving diagnostic accuracy for general medical applications.

PreceptorAI, CARIVA — AI Consultant 2023-Present

- Developed machine learning models in collaboration with Siriraj Radiology Department:
 - Multi-label segmentation model (nnUNet, 0.98 Dice) for vertebrae L3 (muscle, fats) quantification in MRI.
 - Vision-Large Language Model to generate diagnostic reports from chest X-ray images (PyTorch).
 - Generative text-to-image model (diffusion model) for synthetic chest X-ray generation (PyTorch, MONAI).
- Led research and development of Gindee, a VLM based nutrition estimator.

Vision and Learning Lab, VISTEC — Research Assistant, under Prof. Supasorn Suwajanakorn 2023-2024

- Investigated object collocation by developing Score Distillation Sampling (SDS) to leverage Stable Diffusion's priors.
- Enhanced local attention masking in CLIP for zero-guidance segmentation.
- Conducted experiments on diffusion model memorization (CIFAR-10).

Facebook, Notification Backend — Intern 2017

- Implemented a pipeline for network effect prediction (classification/regression) using HiveQL and FB Learner Flow.
- Increased comment metrics by 0.5% in production (2B active users).

MIT Media Lab, Human Dynamics — Undergrad Researcher, under Yan Leng 2018-2020

- Developed an optimization framework (Python, Pandas) for learning network structure and marginal benefits from large-scale datasets (e.g., 400GB+ CDRs).

publications

LUSD: Localized Update Score Distillation for Text-Guided Image Editing

Worameth Chinchuthakun*, **Tossaporn Saengja***, Nontawat Tritrong, Pitchaporn Rewatbowornwong, Pramook Khungurn, Supasorn Suwajanakorn. [arXiv:2503.11054](#) (co-first, under review)

ZeroGuideSeg++: Improving Local Attention Masking in CLIP for Zero-Guidance Segmentation

Pitchaporn Rewatbowornwong, , Nattanat Chatthee, **Tossaporn Saengja**, Ekapol Chuangsuwanich, Supasorn Suwajanakorn. *(under review)*

GPU Poor's Guide to Radiology Report Generation

Kiartnarin Udomlaksakul, et al. (**Tossaporn Saengja** as third author). [Proc. BioNLP 2024](#).

selected academic projects

Accurate Clinical Decision Support Systems (MIT 6.S897: ML for Healthcare, 2017)

Analyzed drivers of clinical decision support dismissals using logistic regression, random forests, LDA.

Paraphrase Identification (MIT 6.867: Machine Learning, 2016)

Designed BoW, word-embedding, SVM, NN, and LSTM models achieving 81.8% F1-score on MSR Paraphrase Corpus.

additional experience

Mahidol Wittayanusorn High School — Computer Science Teacher 2020-2023

- Coached students to represent Thailand in IOI 2022.
- Lectured data science (Python), web dev, blockchain, Computer Olympiad (C++), programming (Python).
- Advised year-long student projects, primarily in machine learning.

Agnos — Technical Cofounder 2019-2023

- Co-developed an AI-powered diagnosis system (Python, Django) used by physicians. (130k+ records, 50k+ active users)
- Led Data Science, AI, and Growth teams, helped securing 30M THB funding (Shark Tank Thailand S3).